Mike Wolak · mike@epromfoundry.com · (512) 944-8586

Amiga AGA Computer Core for the Pano Logic G2 Revision C

An A1200-class Amiga in a Spartan-6 XC6SLX100: the TG68K 68020 with a 2-way cache, the Minimig AGA chipset, chip and fast memory in DDR2, Gayle IDE and four floppy drives on images from a USB stick, Toccata sound, an X-Surf 100 network board and a serial FIFO board, run by a RISC-V firmware with a setup screen. Written to the unit's main flash slot over the network by tnpflash.

HIGHLIGHTS

- TG68K 68020, 4+4 KB cache; OCS, ECS or AGA.
- Floppy and hard-disk images on a USB stick; USBStick: volume.
- USB keyboard, mouse, two DualShock pads.
- Paula and Toccata at 48 kHz; AHI driver.
- X-Surf 100 and PanoTCP networking.
- Setup screen: six pages, every AMIGA.INI key.

KEY PARAMETERS

PARAMETER	VALUE
Board	Pano Logic G2, revision C
FPGA	XC6SLX100-2FGG484
Fabric	0x0AB5EF2D
Firmware	0xBFCDDBD7
Core clock	28.636364 MHz NTSC
Memory	2 × 64 MB DDR2
Flash slot	0x380000, 3,389,656 bytes
Kickstart	supplied by the user

1 MACHINE CONFIGURATIONS

PROFILE	CPU	CHIPSET	CHIP	SLOW	FAST	KICKSTART
A500 OCS	68000	OCS	512 KB	512 KB	none	KICK.ROM
A500+ ECS	68000	ECS	1 MB	none	none	KICK.ROM
ECS 2 MB	68000	ECS	2 MB	512 KB	none	KICK.ROM
A1200 AGA	68020	AGA	2 MB	none	8 MB	KICK_A1200.ROM

Every item is also set on its own in setup or AMIGA.INI. SysInfo 4 reports the A1200 profile's 68020 at 41.60 MHz equivalent.

2 IMPLEMENTATION

RESOURCE	USED	AVAILABLE	SHARE
Occupied slices	13,730	15,822	86 %
Slice LUTs	39,796	63,288	62 %
Slice registers	25,023	126,576	19 %
RAMB16BWER	176	268	65 %
RAMB8BWER	46	536	8 %
DSP48A1	34	180	18 %

ISE 14.7, G2_AMIGA_AGA, PAR cost table 7; all constraints met, worst setup slack 0.334 ns (clk_sys, 20 ns).

3 MEASURED FIGURES

ITEM	RESULT
CPU loop in fast RAM	254 ns an iteration with the cache, 463 ns without
HD0 through scsi.device	609 KB/s read, 620 KB/s write
Disk writes	HD0: 0 of 400 sectors lost; DF0: a 204,800-byte copy reads back equal
PanoTCP	329 KB/s receive, 375 KB/s send; FTP 2,800,640 bytes in 14.9 s
Serial FIFO board	0 lost to 500,000 baud
MIDI loopback	0 lost, 0 errors at 100 % load
Terminal	11,842 bytes/s at 80 × 25 with smooth scrolling

4 INSTALLATION IN BRIEF

1. Unpack the stick archive on a FAT32 or exFAT stick; add a Kickstart image and disk images to \amiga.
2. `tnpflash discover`, `tnpflash info <ip>`, `tnpflash backup <ip> g2-<mac>.bin`.
3. `tnpflash g2write <ip> g2-<mac>.bin pano-g2-amiga-1.0-slot.bin` (about 700 s), then power-cycle.
4. F12 opens setup; `Execute G2Tools:Install` installs the AmigaOS programs.

The full procedure: G2_AMIGA_SETUP_GUIDE.pdf.

5 RELEASE FILES

FILE	BYTES	SHA256 (FIRST 16)	CONTENT
AMIGA.FW	316,164	9ac6c86c94983aff	The firmware, ID 0xBFCDDBD7 (\amiga on the stick)
G2TOOLS.adf	901,120	226072268f875984	DD floppy image: AmigaOS programs and Install
pano-g2-amiga-1.0-amiga-programs.zip	396,896	c36ae94aabb95ca4	The AmigaOS programs as files
pano-g2-amiga-1.0-fabric.bin	3,145,728	9d2e0d695944aa97	The fabric alone (ISE bitgen, SPI order), build ID 0x0AB5EF2D
pano-g2-amiga-1.0-loader.img	243,928	d3164a96473a9471	The loader (flash 0x680000)
pano-g2-amiga-1.0-slot.bin	3,389,656	550b09fee926ff01	Main flash slot at 0x380000: fabric and loader (tnpflash g2write)
pano-g2-amiga-1.0-stick.zip	572,845	bb5a4570ed3bd084	The stick's \amiga directory
tnpflash-linux-arm64	781,928	56c69093f6bdeb75	tnpflash, Linux arm64, static
tnpflash-linux-x86_64	959,736	729661b367a8a523	tnpflash, Linux x86_64, static
tnpflash-macos	153,440	0b35fc09c08727ce	tnpflash, macOS universal
tnpflash-windows.exe	181,995	28988b3c9f9aec44	tnpflash, Windows 32-bit

Full sums in SHA256SUMS of the release. The fabric and the loader are those of the earlier internal release (fabric ab5ef2d, loader 1caf0a5); AMIGA.FW and the AmigaOS programs are new in this release.

6 LIMITS AND UNTESTED ITEMS

ITEM	STATE
FPU	None
31 kHz Amiga modes, NATIVE output	Untested on hardware
Return to the stock main slot	Untested